

Zhuocheng (Tom) Qiao

AI Agent Engineer / AI Backend Engineer

Shanghai | M.S., Signal and Information Processing, East China Normal University (2022)

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Engineering Portfolio: Agent Runtime, AI Harness, Open Source & Project Deep Dives



SUMMARY

Software Engineer II at Microsoft with 4+ years of experience building AI agent runtimes, agent harnesses, backend platforms, and production-grade distributed systems. Experienced in technical design, core implementation, cross-team delivery, and operation of enterprise AI products.

EXPERIENCE

Microsoft | Software Engineer II | Jun 2022-Present

Office Agent (Societas) | General-Purpose Agent Platform | Jul 2025-Feb 2026

A general-purpose agent for Microsoft 365 Personal and Family subscribers, reaching **27K WAU**—approximately 8.5% of eligible users—within two months of global launch; focused on agent runtime, productionization, reliability, and compliance.

- **Native Tool Calling**: Led the migration from text-based XML tool invocation to native function calling, introducing dual-mode rollout and rollback, while extending support across the complete sub-agent tree for reliable, cross-model execution.
- **Reliable Message Delivery**: Independently rebuilt long-running agent messaging with Redis Lists for 24-hour replayable state and Service Bus Sessions for retention, acknowledgement, redelivery, and per-session FIFO; introduced a rollout/rollback transport abstraction for backend interruptions and browser reconnect replay.
- **Agent Proxy**: Designed and delivered Agent Proxy to production from scratch, covering OBO authorization, task persistence and restart recovery, containerization, and CI/CD, enabling reliable external integration as a standalone service.
- **Data Security & Compliance**: Designed the user-level encryption architecture and protocol—hierarchical keys, hashed lookup, self-describing ciphertext, rotation, and gradual migration—adopted across partner modules; independently migrated Societas backend node-to-node traffic from HTTP to HTTPS through Key Vault certificate injection, PFX-to-PEM conversion, server certificate loading, and environment-wide cutover, and added CA verification to Agent Proxy downstream traffic, protecting data at rest and in transit.

Microsoft Copilot | Graph-based Agentic Loop & Production Engineering | Mar 2026-Present

Work spans Epichan's graph-based Harness and Loop Engineering practices, and Studio's production AI product and delivery systems.

- **Epichan Graph + Loop Engineering**: Built on hands-on experience closing **64 Sev2 accessibility defects** in Studio in two months to codify three recurring failure modes as validation rules; designed and iterated a 11y-remediation-workflow on top of Epichan's existing graph-based Harness, enabling Developer, Reviewer, and Judge to reproduce, fix, and independently validate against shared evidence standards.
- **Studio Engineering & Production Reliability**: Hands-on understanding of Copilot's inference and message-orchestration architecture, spanning session ingress, Turn/Orchestrator scheduling, model-tool loops, safety/citation processing, and streaming responses; contributed core CI speed and stability improvements, including change classification, test parallelization, and automated flaky detection, isolation, and tracking, helping reduce team build time by about 30%; designed and enabled the hotfix-to-production path from scratch, optimized artifact promotion, and supported multi-region progressive delivery and on-call operations, using release history, telemetry, and regional health to rapidly diagnose incidents and apply tiered containment and recovery.

Office Plus | Commercial Backend Platform | Jul 2024-Jun 2025

Multi-service commercialization platform built with .NET/ABP and Azure, spanning orders, payments, entitlements, identity, and third-party integrations.

- **Plagiarism-Checking Commercialization**: Led end-to-end productization across file upload, orders, payments, entitlements, asynchronous processing, result delivery, retries, and automatic refunds; processed **6,320 orders** in the first **10 days**.
- **Identity & Session Governance**: Led account soft deletion, refresh-token modernization, and device-session governance, reducing record growth from **MAU x devices x 30 to MAU x devices** while supporting quotas, revocation, and gradual migration.

Earlier Projects: XPay (Jun-Oct 2022) | PC Manager Client (Nov 2022-Jun 2024)

TECHNICAL SKILLS

Agent Engineering: Function Calling, Multi-Agent/Sub-Agent, Agent Harnesses, Skills, Tool Orchestration, Context & Memory, Automated Evaluation & Quality Gates

Backend: Python, FastAPI, C#/ .NET, Java, REST/SSE, Distributed Tasks & Asynchronous Processing

Infrastructure: Azure, Service Bus, Redis, PostgreSQL, Docker, CI/CD, Logging & Observability

OPEN-SOURCE CONTRIBUTIONS

Contributed to popular open-source projects including Hermes Agent and Multica, with **10+ fixes** merged or incorporated upstream across agent security, runtime reliability, tenant isolation, and API robustness; experienced in issue discovery, root-cause analysis, patches, tests, and maintainer collaboration.